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(54) DEVICE FOR INDUCTIVE ENERGY TRANSMISSION IN A HUMAN BODY AND USE OF THE DEVICE

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(58) **Field of Classification Search**

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See application file for complete search history.

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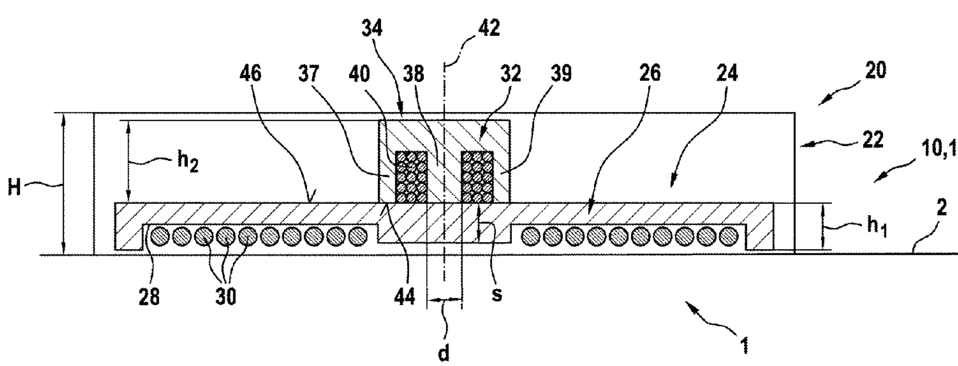
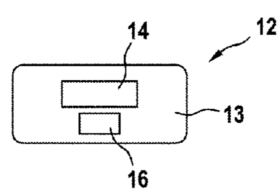
(57) **ABSTRACT**

The invention relates to a device (**10**) for inductive energy transmission into a human body (**1**), having a transmitter coil (**24**) and/or a receiver coil (**14**) having a first magnetic core (**26**) and a resonance or choke coil (**16**, **34**) having a second magnetic core (**32**), wherein the first magnetic core (**26**) forms a part of the second magnetic core (**32**).

18 Claims, 1 Drawing Sheet





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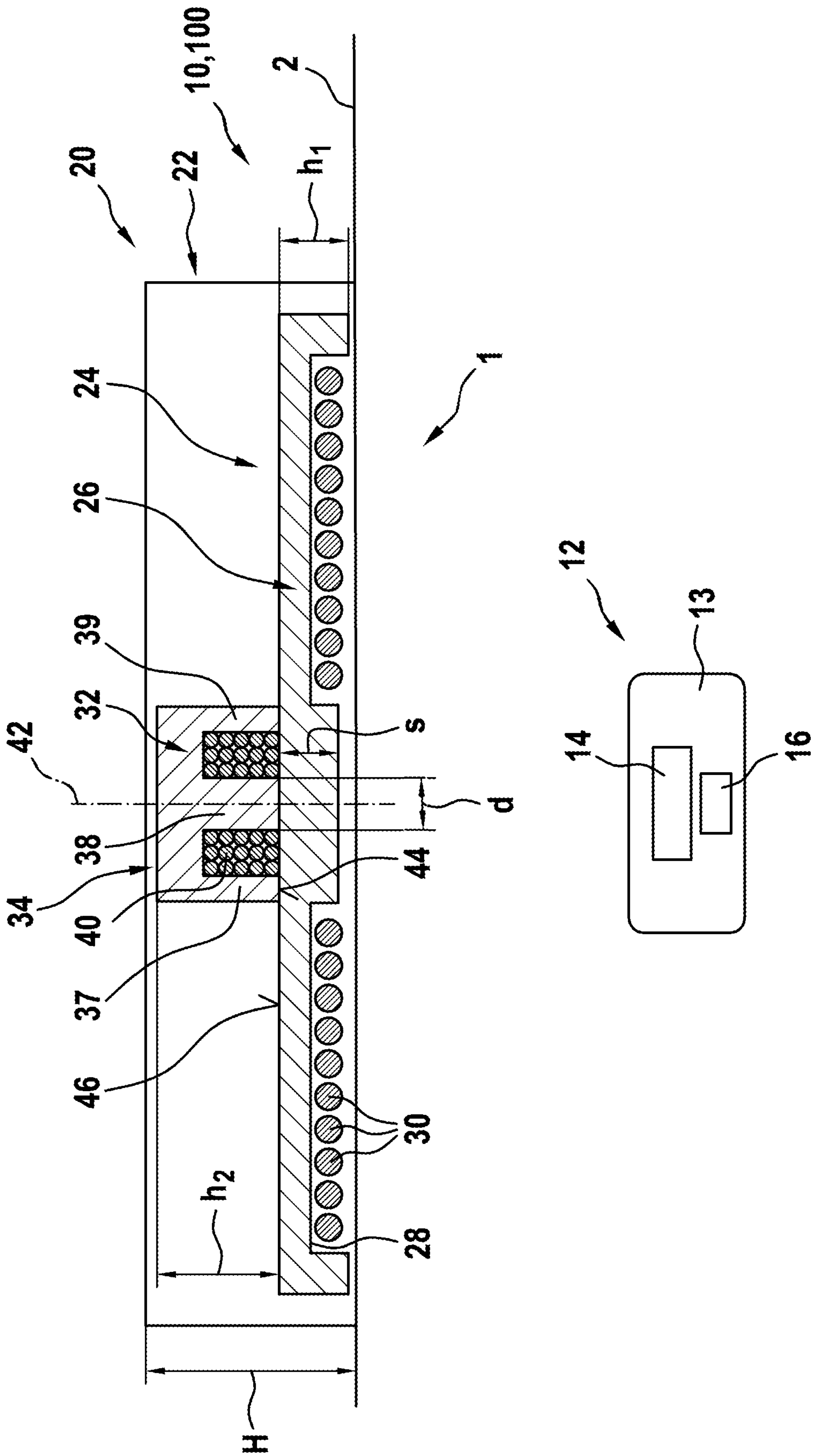
10,100
2
20
22
h_1
1
24
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12
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s
42
14
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d
38
44
34
40
37
46
30
h_2
28
H

DEVICE FOR INDUCTIVE ENERGY TRANSMISSION IN A HUMAN BODY AND USE OF THE DEVICE

BACKGROUND

Field

The invention relates to a device for inductive energy transmission into a human body. The invention further relates to the use of a device according to the invention.

Medical technology uses devices for inductive energy transmission, in which a transmitter unit disposed outside a human body comprises a transmitter coil, the magnetic field of which induces energy in the receiver coil of a receiver unit disposed [in] the human body, which at least indirectly serves to charge a battery disposed in the human body. Such a system is known as a so-called VAD (ventricular assist device) system used to operate a pump that supports the heart of a patient (DE 10 2016 106 683 A1).

For such a device, and this applies to both the receiver unit and the transmitter unit, it is essential that it is configured in as compact a manner as possible. On the one hand, in terms of the receiver unit, this minimizes the space required in the body for the receiver unit, while on the other hand, in terms of the transmitter unit, it increases the wearing comfort of such a transmitter unit under the patient's clothing. The transmitter unit or the receiver unit furthermore typically comprise a housing, in which the components of the respective unit are disposed. It is also known that, in addition to the transmitter coil or the receiver coil, an electronic circuit is required for the operation of said coils, which typically also comprises a so-called resonance or choke coil. Like the transmitter coil or the receiver coil, such a resonance or choke coil comprises a magnetic core and a wire winding that interacts with the magnetic core. In the case of the resonance or choke coil, this is typically a component of a resonance circuit and, in the state of the art, is always disposed discretely, i.e. separately from the transmitter coil or receiver coil, on a circuit carrier (circuit board) together with other electronic components. The separate arrangement and configuration of the magnetic cores for the transmitter coil or receiver coil and for the resonance or choke coil results in an increase of the overall height of the transmitter unit or receiver unit, in particular with regard to the respective magnetic cores, because the resonance or choke coil is disposed above or below the level of the transmitter coil or receiver coil and thus also affects the overall height of the housing of the transmitter unit or receiver unit.

DISCLOSURE OF THE INVENTION

Summary

The device according to the invention for inductive energy transmission having the features of Claim **1** has the advantage that it enables a particularly compact arrangement of the transmitter coil or receiver coil and the resonance or choke coil in the transmitter unit or the receiver unit.

The invention is based on the idea of using or configuring a part of the magnetic core of the transmitter coil or the receiver coil simultaneously as a part of the magnetic core of the resonance or choke coil or vice versa. This not only saves material for the magnetic cores, but also enables a particularly compact arrangement of the two magnetic cores relative to one another.

Advantageous further developments of the device for inductive energy transmission into a human body according to the invention are presented in the subclaims.

In a first structural configuration of the device, it is provided that the first magnetic core (of the transmitter or receiver coil) is disposed in direct contact with an end face of the second magnetic core (of the resonance or choke coil).

In order to fulfill the intended functionality, it is furthermore provided in a preferred structural configuration that the transmitter coil or the receiver coil comprises wire windings which are disposed concentrically to one another and the resonance or choke coil comprises second wire windings which are disposed concentrically to one another, and that the first wire windings are disposed radially outside the second wire windings.

The last-mentioned configuration in particular leads to a geometric arrangement, in which the resonance or choke coil is disposed concentrically to a longitudinal axis of the first magnetic core.

A structural arrangement or configuration of the two coils (transmitter coil and receiver coil or resonance or choke coil) as described thus far in particular enables a desired decoupling of the transmitter and the receiver coil with the resonance or choke coil due to both the selected topology and the design of the transmitter or receiver coil itself. The transmitter coil and the receiver coil in particular have a very low coupling factor to the resonance or choke coil, because the (magnetic) fields of the transmitter coil or receiver coil are enclosed on one side by the ferrite core. The return of the magnetic fields on the other side then takes place via the air over a longer distance and thus leads to the aforementioned low coupling. The structural arrangement of the coils as described thus far typically has a very low coupling factor of typically up to about 0.05.

Another preferred structural configuration provides that the first magnetic core is disc-shaped and has at least one groove-like recess which extends radially around the longitudinal axis for receiving first wire windings of the transmitter or receiver coil.

In particular in the last-mentioned design, it is further provided that the resonance or choke coil is disposed on the side of the at least one recess which faces away from the first wire windings.

To positively affect the saturation of the core material of the magnetic core of the resonance or choke coil and for the purpose of optimal guidance of the magnetic field lines, it is also advantageous if, in the region of contact with the second magnetic core, the first magnetic core has a wall thickness which corresponds to the wall thickness of a middle leg of the E-core of the second magnetic core.

The invention further also includes the use of a device for energy transmission into a human body according to the invention as described thus far, in particular as a component of a VAD system.

Further advantages, features and details of the invention emerge from the following description of preferred design examples and with reference to the drawing.

BRIEF DESCRIPTION OF THE DRAWING

The only FIGURE shows a schematic illustration of a device for inductive energy transmission into a human body.

DETAILED DESCRIPTION

The device **10** shown in the only FIGURE is in particular configured as a component of a so-called VAD system **100**, wherein the VAD system **100** comprises a pump for supporting the heart function of a patient that is operated or supplied with energy via a not-depicted battery. This battery is charged by means of the device **10**.

The device **10** comprises a receiver unit **12** that is disposed in a human body **1** and comprises a housing **13**, in which a receiver coil **14**, which is shown in the illustration of the FIGURE in a highly simplified manner, is disposed. The receiver coil **14** is used for at least indirect charging of said battery, for example via not-depicted wire connections to the battery and a not-depicted electronic circuit. In addition to the receiver coil **14**, the receiver unit **12** further comprises a choke coil **16**, which is likewise shown purely schematically and is not a component of a receiving oscillator circuit.

The receiver unit **12** interacts with a transmitter unit **20** disposed outside the body **1**. As an example, the transmitter unit **20** comprises a housing **22** which, on the one hand, can be positioned in at least indirect contact with the skin **2** of the body **1** and in alignment with the receiver unit **12** and, on the other hand, comprises a transmitter coil **24** in its interior. The transmitter coil **24** consists of a disc-shaped first magnetic core **26**, which comprises a radially circumferential, groove-like recess **28** on the side facing the receiver unit **12**. First wire windings **30**, which are disposed concentrically to one another around a longitudinal axis **42** of the first magnetic core **26**, are disposed in the region of the recess **28** and serve to produce a not-depicted magnetic field in order to induce electrical energy in the receiver coil **12** which is used to charge said battery.

As an example, the transmitter coil **24** has a height h_1. On the side of the first magnetic core **26** facing away from the receiver unit **12**, a second magnetic core **32** is disposed as part of a resonance or choke coil **34**. The second magnetic core **32** is configured in the form of an E-magnetic core **32**. Second wire windings **40** of the resonance or choke coil **34** are concentrically wound around a (central) middle leg **38** of the second magnetic core **36**.

As can be seen from the illustration of the FIGURE, the middle leg **38** is also disposed concentrically to the longitudinal axis **42** of the first magnetic core **26**. The second wire windings **40** furthermore extend radially inside the first wire windings **30** of the transmitter coil **24**.

With its three legs **37** to **39**, the end face **44** of the second magnetic core **36** facing the first magnetic core **26** directly abuts the end face **46** of the first magnetic core **26**. The illustration of the FIGURE also shows that, in the region in which the second magnetic core **32** and the first magnetic core **26** overlap, the thickness d of the middle leg **38** of the second magnetic core **32** corresponds to the wall thickness s of the first magnetic core **26**. The second magnetic core **32** has a height h_2. The illustration of the FIGURE further shows that the total height H of the housing **22** of the transmitter unit **20** substantially depends on or is determined by the sum of the two heights h_1 and h_2 of the two magnetic cores **26** and **32**.

The device **10** as described thus far can be changed or modified in many ways without departing from the idea of the invention. It is in particular noted that the specific configuration of the transmitter coil **24** and the resonance or choke coil **34**, or the magnetic cores **26**, **32** thereof, has been described on the basis of the transmitter unit **20**. It is just as possible for the described structure to be provided in the same way in the region of the receiver unit **12**. This likewise results in a minimization of the overall height of the housing **13** of the receiver unit **12**.

What is claimed is:

1. A device for inductively transmitting energy into or receiving energy within a human body, the device comprising:

a first coil including a first magnetic core, wherein the first magnetic core has an end face and at least one leg extending away from a side opposite the end face and is configured to receive a plurality of wire windings opposite the end face and around the at least one leg, and

a second coil including a second magnetic core, the second magnetic core comprising three legs abutting the end face of the first magnetic core, wherein the second coil includes at least one recess extending radially around a middle leg of the three legs, the at least one recess configured to receive a plurality of wire windings,

wherein the first magnetic core and the second magnetic core have E-shaped cross-sections facing in a same direction, and wherein a width of the first magnetic core disposed in direct contact with the second magnetic core corresponds to a width of the middle leg of the second magnetic core.

2. The device according to claim **1**, wherein the plurality of wire windings of the second coil wind concentrically around the middle leg of the second magnetic core.

3. The device according to claim **1**, wherein the first coil comprises a plurality of first wire windings disposed concentrically relative to one another, wherein the second coil comprises a plurality of second wire windings disposed concentrically relative to one another, and wherein the plurality of first wire windings are disposed radially outside the plurality of second wire windings.

4. The device according to claim **1**, wherein the second coil is disposed concentrically to a longitudinal axis of the first magnetic core.

5. The device according to claim **1**, wherein the first magnetic core is disc-shaped.

6. The device according to claim **5**, wherein the first magnetic core comprises a radially circumferential recess.

7. The device according to claim **1**, further comprising a housing surrounding the first coil and the second coil, a height of the housing being equal to a height of the first coil and a height of the second coil.

8. A method for inductively transmitting energy, the method comprising:

inductively transmitting energy using a transmitter device and a receiver device, wherein at least one of the transmitter device and the receiver device comprises:

a first coil including a first magnetic core, wherein the first magnetic core has an end face and at least one leg extending away from a side opposite the end face and is configured to receive a plurality of wire windings opposite the end face and around the at least one leg, and

a second coil including a second magnetic core, the second magnetic core comprising three legs abutting the end face of the first magnetic core, wherein the second coil includes at least one recess extending radially around a middle leg of the three legs, the at least one recess configured to receive a plurality of wire windings,

wherein the first magnetic core and the second magnetic core have E-shaped cross-sections facing in a same direction, and wherein a width of the first magnetic core disposed in direct contact with the second magnetic core corresponds to a width of the middle leg of the second magnetic core.

9. The method of claim **8**, wherein at least one of the transmitter device and the receiver device further comprises a housing, wherein the housing is in at least indirect contact with skin of a human body.

10. The method of claim **8**, wherein the transmitter device inductively transmits energy and the receiver device inductively receives energy.

11. The method of claim **10**, wherein the energy received at the receiver device is used for charging a battery in communication with the receiver device.

12. A system for inductive energy transmission, the system comprising:

a first device including a first coil and a second coil; and

a second device,

wherein the first coil includes a first magnetic core, wherein the first magnetic core has an end face and at least one leg extending away from a side opposite the end face and is configured to receive a plurality of wire windings opposite the end face and around the at least one leg and wherein the second coil includes a second magnetic core,

wherein the second magnetic core comprises three legs abutting the end face of the first magnetic core, wherein the second coil includes at least one recess extending radially around a middle leg of the three legs, the at least one recess configured to receive a plurality of wire windings,

wherein the first magnetic core and the second magnetic core have E-shaped cross-sections facing in a same direction and

wherein the first and second devices are disposed in alignment on opposite sides of skin of a body to transfer energy into the body, and wherein a width of the first magnetic core disposed in direct contact with the second magnetic core corresponds to a width of the middle leg of the second magnetic core.

13. The system of claim **12**, wherein the plurality of wire windings of the second coil wind concentrically around the middle leg of the second magnetic core.

14. The system of claim **12**, wherein the first coil comprises a plurality of first wire windings disposed concentrically relative to one another, wherein the second coil comprises a plurality of second wire windings disposed concentrically relative to one another, and wherein the plurality of first wire windings are disposed radially outside the plurality of second wire windings.

15. The system of claim **12**, wherein the second coil is disposed concentrically to a longitudinal axis of the first magnetic core.

16. The system according to claim **12**, further comprising a housing surrounding the first coil and the second coil, a height of the housing equal to a height of the first coil and a height of the second coil.

17. The system according to claim **12**, wherein the first magnetic core is disc shaped.

18. The system according to claim **17**, wherein the first magnetic core comprises a radially circumferential recess.

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